

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		permanent dipoles between molecules in ICl but Cl ₂ has only temporary dipoles / van der Waals forces between molecules (1) permanent dipoles stronger than temporary dipoles / van der Waals forces (1) or ICl larger molecule than Cl ₂ so more electrons (1) more van der Waals forces between molecules (1)	2			2		
		(ii)		octahedral	1			1		
		(iii)		[ClF ₂] ⁺ has 2 bond pairs and 2 lone pairs (1) [ClF ₂] ⁻ has 2 bond pairs and 3 lone pairs (1) student is correct - different numbers of electron pairs will give rise to different shapes (1)			3	3		
				Question 13 total	6	3	5	14	2	4